

GEORGIA INSTITUTE OF TECHNOLOGY
COE3001 MECHANICS OF DEFORMABLE BODIES

Final exam - 2026 Spring

Name: _____ GTID: _____

Please read the following instructions

- ✧ You have **170** minutes to complete this exam.
- ✧ You may use a single, double-sided, letter sized formula sheet.
- ✧ This question booklet contains **6** questions, **13** pages (including the cover) for a total of **40 points + 3 bonus**. Check to see if any pages are missing.
- ✧ The maximum achievable score is **40**, including any bonus points.
- ✧ DO NOT scribble or do rough work or make any stray marks on it. Use separate sheet for rough work.
- ✧ All the questions are compulsory, and all the notations have their usual meaning.
- ✧ Show all necessary steps for full credit. Clearly indicate your final answers.
- ✧ Partial credit will be given for showing your work
- ✧

Section 1 – Physical Intuition and Conceptual Reasoning (15 pts)

Answer all questions. Unless explicitly requested, lengthy calculations are not required. Credit will be based on the correctness of the answer and the quality of the mechanical reasoning. Where appropriate, support your answer with a brief sketch, limiting case, or physical argument.

Problem 1 – Foundational questions

1.1 Bending stiffness and cross-sectional orientation (0.5+1+1.5 pts)

A prismatic beam with a rectangular cross section of width b and height h ($h > b$) is subjected to transverse bending moments in two different orientations:

- Case A: the dimension h is vertical
- Case B: the beam is rotated by 90° , so the dimension b is vertical

Assume the material, beam length, and moment are the same in both cases.

- (a) Which case has the smaller deflection? Case A or Case B
- (b) Which case develops the smaller bending stress at the outer surface under the same bending moment?
Case A or Case B
- (c) Explain physically why changing the orientation has such a large effect.

1.2 Stress transformation under pure shear (0.5+0.5+1 pts)

Under plane stress assumption, a material element experiences a pure shear stress state:

$$\tau_{xy} \neq 0, \quad \sigma_x = \sigma_y = 0$$

- (a) Are the principal stresses zero or nonzero?
- (b) On planes rotated 45° from the original faces, is the stress state still pure shear?
- (c) Describe all the possible stress states for this material element

Problem 2 – Counterintuitive mechanics and higher-insight questions

These questions are designed to test whether you can distinguish between **stress, strain, displacement, stiffness, strength, and stability**. Many of the statements below sound plausible at first glance. Read carefully.

2.1 True / False without Justification (6*0.5 pts)

For each statement, indicate whether it is True or False, **no justification needed or graded**.

- (a) A point in a structure can have zero strain while the displacement at that point is nonzero.
- (b) If the external load on a member is zero, then the internal stress in the member must also be zero.
- (c) Two beams with the same material and the same cross-sectional area must have similar resistance to bending.
- (d) A cross section where the internal force is zero must be stress-free.
- (e) Slenderness can affect failure even when the material and cross-sectional area are unchanged.
- (f) Changing the orientation of the stress element changes the description of the stress state, but not the physical stress state itself..

2.1 True / False with Justification (7*1 pts)

For each statement, indicate whether it is True or False, and justify your answer briefly but precisely. **A correct choice without justification will receive limited credit.**

- (a) If the internal bending moment at a section is zero, then the section cannot fail in a normal-stress mode.
- (b) A member with a larger elastic modulus is always less likely to fail.
- (c) If two columns have the same material and the same length, then the one with the larger cross-sectional area must have the higher critical buckling load.
- (d) A point in a body may have zero stress in one chosen direction while still being under a nonzero overall stress state.
- (e) If a structure shows very small deflection under load, then the internal stress everywhere must also be

small.

- (f) A member can change shape without changing its volume significantly.
- (g) If the net axial force on a cross section is zero, then the normal stress on that section must be zero everywhere.

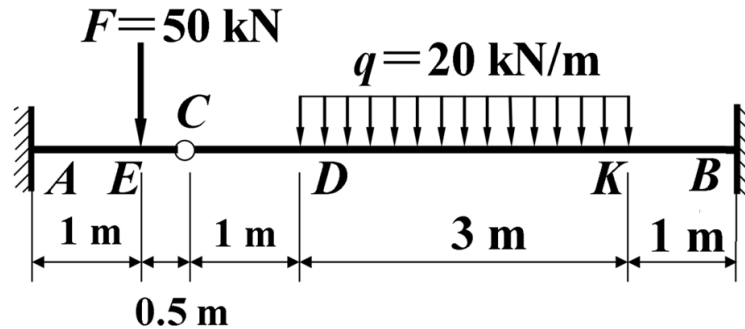
Problem 3 (6 pts)

Beams AC and BC are pin-connected at C, with ends A and B fixed. Both beams have the same flexural rigidity EI . The applied concentrated force is $F=50$ kN, and the uniformly distributed load is $q=20$ kN/m.

(a) calculates the vertical reaction forces and reaction bending moments on A and B ends.

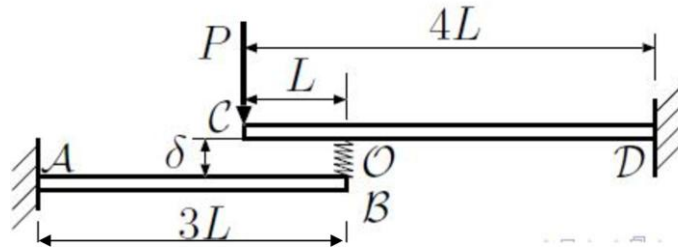
(b) Draw the SFD and BMD for the beam.

(Hint: There are no horizontal reactions (loading is vertical only). The structure is statically indeterminate.)



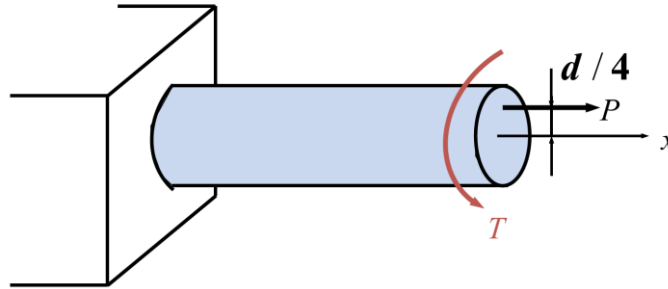
Problem 4 (7 points + 3 bonus)

As shown below, two prismatic cantilever beams AB and CD, both having Young's modulus E and moment of inertia I , are fixed at A and D, respectively. On beam AB, a spring with a constant k is fixed to the free end B. Initially with no system loading, the spring has a length of δ (the distance between C and beam AB, $\delta \ll L$) and its tip touches the beam CD at O with no force. Now a vertical force P is applied downward at C on beam CD. Determine the relation between δ and the magnitude of P . Assume the horizontal displacement can be neglected. Also, for simplicity, take $EI = 9kL^3$ and express the result in terms of k and L .



Problem 5 (6 points)

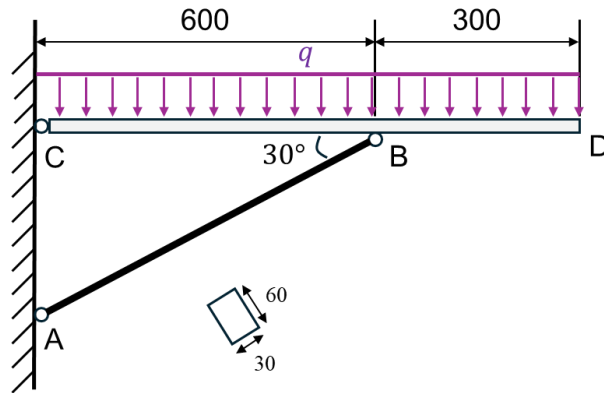
As shown below, for a solid circular shaft, the left end is fixed, and the free end is subject to an external concentrated load with magnitude of P at $d/4$ above the center. The total length of the shaft is L , and the diameter of the wheel is d . Determine the most dangerous point(s) according to the Maximum shear stress theory (Hint: $\sigma_{r3} = \sigma_{\max} - \sigma_{\min}$). What is the maximum principal stress σ_1 at the dangerous point(s)?



Problem 6 (6 points)

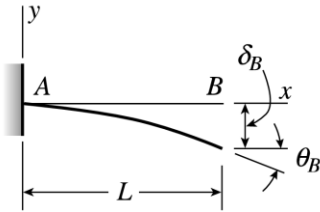
A solid circular beam CD has a diameter of $d = 100$ mm, $[\sigma] = 160$ MPa. Bar AB with a rectangular cross section, $E = 200$ GPa, $\sigma_{pl} = 200$ MPa, $\sigma_s = 240$ MPa. The linear relation of inelastic buckling has coefficient $a = 310$ MPa and $b = 1.14$ MPa, The safety factor of stability is $n_{st} = 3$. Ignore the transverse shear and axial force in beam CD. Please determine the allowable load q_{max} .

(Hint: the force acting on the bar AB is along its longitudinal axis)



(unit: mm)

Deflection table



Notation:

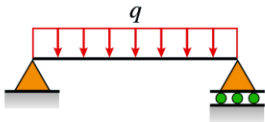
v = deflection in the y direction (positive upward)

$v' = dv/dx$ = slope of the deflection curve

$\delta_B = -v(L)$ = deflection at end B of the beam (positive downward)

$\theta_B = -v'(L)$ = angle of rotation at end B of the beam (positive clockwise)

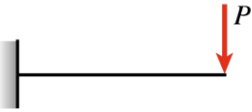
EI = constant



$$v = -\frac{qx}{24EI}(L^3 - 2Lx^2 + x^3)$$

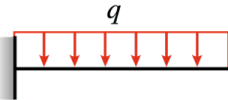
$$v' = -\frac{q}{24EI}(L^3 - 6Lx^2 + 4x^3)$$

$$\delta_C = \delta_{\max} = \frac{5qL^4}{384EI} \quad \theta_A = \theta_B = \frac{qL^3}{24EI}$$



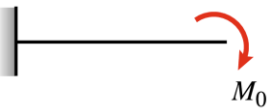
$$v = -\frac{Px^2}{6EI}(3L - x) \quad v' = -\frac{Px}{2EI}(2L - x)$$

$$\delta_B = \frac{PL^3}{3EI} \quad \theta_B = \frac{PL^2}{2EI}$$



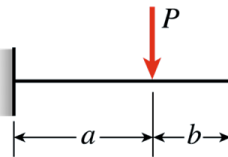
$$v = -\frac{qx^2}{24EI}(6L^2 - 4Lx + x^2) \quad v' = \frac{qx}{6EI}(3L^2 - 3Lx + x^2)$$

$$\delta_B = \frac{qL^4}{8EI} \quad \theta_B = \frac{qL^3}{6EI}$$



$$v = -\frac{M_0x^2}{2EI} \quad v' = -\frac{M_0x}{EI}$$

$$\delta_B = \frac{M_0L^2}{2EI} \quad \theta_B = \frac{M_0L}{EI}$$



$$v = -\frac{Px^2}{6EI}(3a - x) \quad v' = -\frac{Px}{2EI}(2a - x) \quad (0 \leq x \leq a)$$

$$v = -\frac{Pa^2}{6EI}(3x - a) \quad v' = -\frac{Pa^2}{2EI} \quad (a \leq x \leq L)$$

$$\text{At } x = a: \quad v = -\frac{Pa^3}{3EI} \quad v' = -\frac{Pa^2}{2EI}$$

$$\delta_B = \frac{Pa^2}{6EI}(3L - a) \quad \theta_B = \frac{Pa^2}{2EI}$$